Integration by partial fractions

Note

- We use partial fractions if the integ. contains a rational fn, s.t. the den. can be factorized.

On condition:

degree of Num. < degree of den.

or o.w.

we use long division first.

Case (i)

$$\frac{------}{(x+a)(x+b)(x+c)} = \frac{A}{(x+a)} + \frac{B}{(x+b)} + \frac{C}{(x+c)}$$

Case (ii)

$$\frac{-------}{(x+a)(mx^2+bx+c)} = \frac{A}{(x+a)} + \frac{Bx+C}{mx^2+bx+c}$$

Case (iii)

$$\frac{----}{(x+a)^3} = \frac{A}{(x+a)} + \frac{B}{(x+a)^2} + \frac{C}{(x+a)^3}$$

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Sheet (2) Section (B)

Integration by Partial fractions

Lec. Prob.

(13) $I = \int \frac{2x^3+x^2+9x+4}{x^2+4}\,dx$

$3 \geq 2$ L.D.

$$I = \int 2x+1 + \frac{x}{x^2+4}\,dx$$

$$= \int 2x+1+\frac{1}{2}\frac{2x}{x^2+4}\,dx$$

$$= 2\frac{x^2}{2}+x+\frac{1}{2}\ln|x^2+4|+C$$

$$\begin{array}{r|l} & 2x+1 \\ \hline x^2+4 & 2x^3+x^2+9x+4 \\ & -\ 2x^3+8x \\ \hline & x^2+x+4 \\ & -\ x^2+4 \\ \hline & x \end{array}$$

(12) $I = \int \frac{x^5+8x^3+16x}{x^2+4}\,dx$

$5 \geq 2$ L.D.

$$I = \int x^3+4x + \cancel{\frac{0}{x^2+4}}\,dx$$

$$= \frac{x^4}{4}+4\frac{x^2}{2}+C$$

$$= \frac{x^4}{4}+2x^2+C$$

$$\begin{array}{r|l} & x^3+4x \\ \hline x^2+4 & x^5+8x^3+16x \\ & -\ x^5+4x^3 \\ \hline & 4x^3+16x \\ & -\ 4x^3+16x \\ \hline & 0 \end{array}$$

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(14) $I = \int \frac{1}{(x-1)(x+1)} dx$

$$* \boxed{\frac{1}{(x-1)(x+1)} = \frac{A}{(x-1)} + \frac{B}{(x+1)}}$$

• Mult. both sides by $(x-1)(x+1)$

$\rightarrow 1 = A(x+1) + B(x-1)$

at $x = 1$

$1 = A(2) + 0 \longrightarrow A = 1/2$

at $x = -1$

$1 = 0 + B(-2) \longrightarrow B = -1/2$

$$\therefore I = \int \frac{1/2}{x-1} + \frac{-1/2}{x+1} \, dx$$

$$= \int 1/2 \frac{1}{x-1} - 1/2 \frac{1}{x+1} \, dx$$

$$= 1/2 \ln|x-1| - 1/2 \ln|x+1| + C$$

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(19) $I = \int \frac{x^2+3x-3}{(x+1)(x^2+4)} \, dx$

$$* \boxed{\frac{x^2+3x-3}{(x+1)(x^2+4)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+4}}$$

• Mult. both sides by $(x+1)(x^2+4)$

$\rightarrow x^2+3x-3 = A(x^2+4)+(x+1)(Bx+C)$

at $x=-1$

$1-3-3 = 5A+0 \longrightarrow A=-1$

at $x=0$

$-3 = 4A + C$

$-3 = 4(-1)+C \longrightarrow C = 1$

at $x=1$

$1+3-3 = 5A+2(B+C)$

$1 = 5(-1)+2(B+1) \longrightarrow B=2$

$\therefore I = \int \frac{-1}{x+1} + \frac{2x+1}{x^2+4} \, dx$

$= \int -\frac{1}{x+1} + \frac{2x}{x^2+4} + \frac{1}{(x)^2+(2)^2} \, dx$

$= -\ln|x+1| + \ln|x^2+4| + \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + C$

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(16) $I = \int \frac{x}{(x-1)(x+1)^2}\, dx$

$$* \quad \boxed{\frac{x}{(x-1)(x+1)^2} = \frac{A}{(x-1)} + \frac{B}{(x+1)} + \frac{C}{(x+1)^2}}$$

Multi both sides by $(x-1)(x+1)^2$

$\rightarrow \quad x = A(x+1)^2 + B(x-1)(x+1) + C(x-1)$

at $x = -1$

$-1 = 0 + 0 + C(-2) \longrightarrow C = \frac{1}{2}$

at $x = 1$

$1 = 4A + 0 + 0 \longrightarrow A = \frac{1}{4}$

at $x = 0$

$0 = A - B - C$

$0 = \frac{1}{4} - B - \frac{1}{2} \longrightarrow B = -\frac{1}{4}$

$$\therefore I = \int \frac{\frac{1}{4}}{(x-1)} + \frac{-\frac{1}{4}}{(x+1)} + \frac{\frac{1}{2}}{(x+1)^2}\, dx$$

$$= \int \frac{1}{4}\frac{1}{(x-1)} - \frac{1}{4}\frac{1}{(x+1)} + \frac{1}{2}(x+1)^{-2}\, dx$$

$$= \frac{1}{4}\ln|x-1| - \frac{1}{4}\ln|x+1| + \frac{1}{2}\frac{(x+1)^{-1}}{-1} + C$$

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(19) $I = \int \frac{dx}{x^3 - 2x^2 + x} = \int \frac{dx}{x(x^2 - 2x + 1)} = \int \frac{dx}{x(x-1)^2}$

$$* \quad \boxed{\frac{1}{x(x-1)^2} = \frac{A}{x} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2}}$$

Mult. both sides by $x(x-1)^2$

$\rightarrow 1 = A(x-1)^2 + Bx(x-1) + Cx$

at $x = 0$

$1 = A + 0 + 0 \longrightarrow A = 1$

at $x = 1$

$1 = 0 + 0 + C \longrightarrow C = 1$

at $x = 2$

$1 = A + 2B + 2C$

$1 = 1 + 2B + 2 \longrightarrow B = -1$

$\therefore I = \int \frac{1}{x} + \frac{-1}{x-1} + \frac{1}{(x-1)^2} \, dx$

$= \int \frac{1}{x} - \frac{1}{x-1} + (x-1)^{-2} \, dx$

$= \ln|x| - \ln|x-1| + \frac{(x-1)^{-1}}{-1} + C$

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(21) $I = \int \frac{x^2+7x+9}{(x+2)(x+1)^3}\,dx$

$$* \quad \frac{x^2+7x+9}{(x+2)(x+1)^3} = \frac{A}{(x+2)} + \frac{B}{(x+1)} + \frac{C}{(x+1)^2} + \frac{D}{(x+1)^3}$$

Mult. both sides by $(x+2)(x+1)^3$

$\rightarrow x^2+7x+9 = A(x+1)^3 + B(x+2)(x+1)^2 + C(x+2)(x+1) + D(x+2)$

at $x=-1$

$3 = 0+0+0+D \longrightarrow D=3$

at $x=-2$

$-1 = -A+0+0+0 \longrightarrow A=1$

at $x=0$

$9 = A+2B+2C+2D$

$9 = 1+2B+2C+6$

$\therefore B+C=1$ ——— eq: ①

at $x=1$

$17 = 8A+12B+6C+3D$

$17 = 8+12B+6C+9$

$\therefore 2B+C=0$ ——— eq: ②

solving eq^ns ① & ② $\longrightarrow B=-1$ & $C=2$

$$I = \int \frac{1}{x+2} - \frac{1}{x+1} + 2(x+1)^{-2} + 3(x+1)^{-3}\,dx$$

$$= \ln|x+2| - \ln|x+1| + 2\frac{(x+1)^{-1}}{-1} + 3\frac{(x+1)^{-2}}{-2} + C$$

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